

SIMPLE WEB APP

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ABOUT

Proxy (Nginx) → Front-End
(served by Nginx) → Back-
End (Flask) → Database
(Postgres)

Simple Web App

Call /api/helloCheck DatabaseLoad Items Table

Add New Item (write to DB):

Add Item

Response:

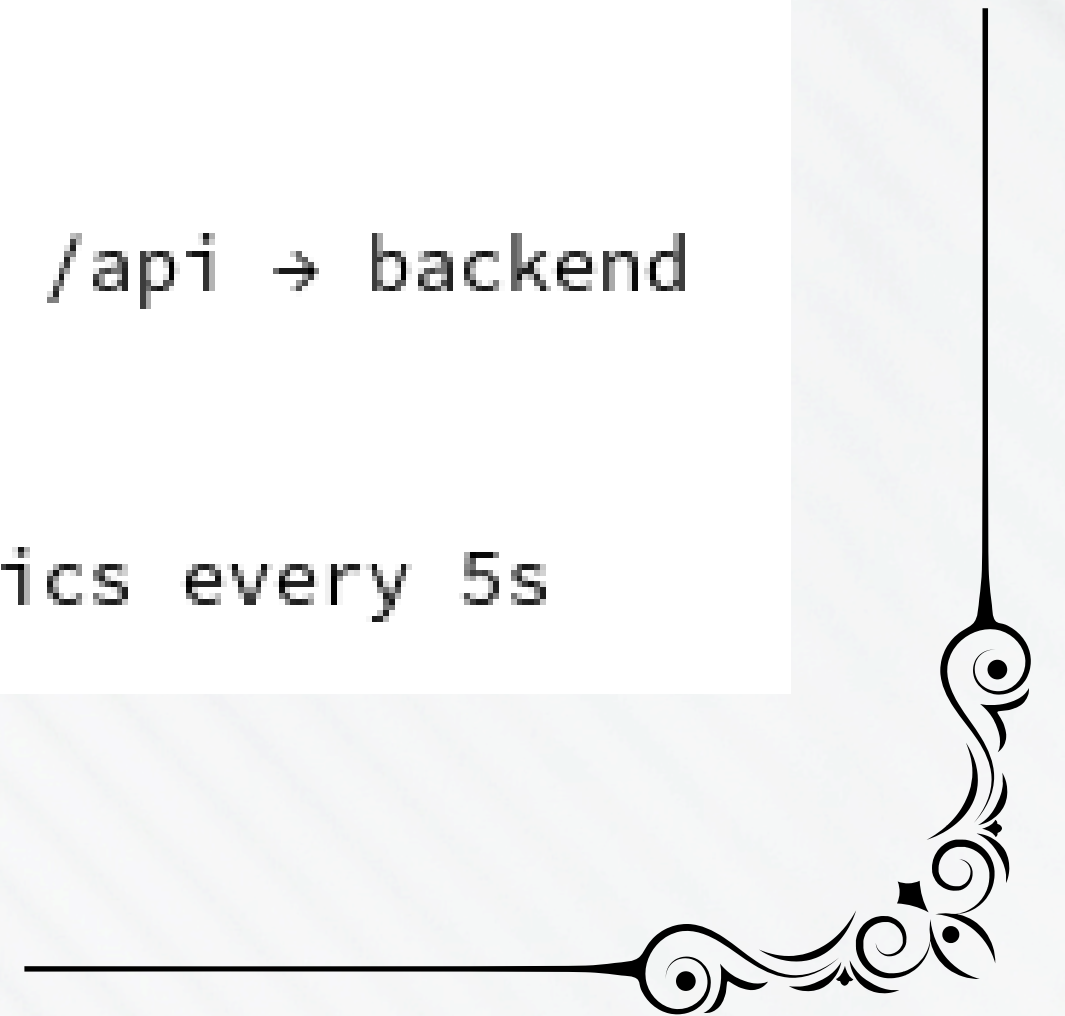
```
{  "message": "Hello from the backend!",  "status": "ok"}
```

Items Table (from Postgres):

ID	Name	Description	Created At
1	Front-End	Static page served by Nginx	2026-07-11T10:18:36.169483
2	Back-End	Flask API service	2026-07-11T10:18:36.169483
3	Database	PostgreSQL, isolated on the backend network	2026-07-11T10:18:36.169483
4	Proxy	Nginx reverse proxy routing / and /api	2026-07-11T10:18:36.169483
5	Monitoring	Prometheus + Grafana	2026-07-11T10:18:36.169483

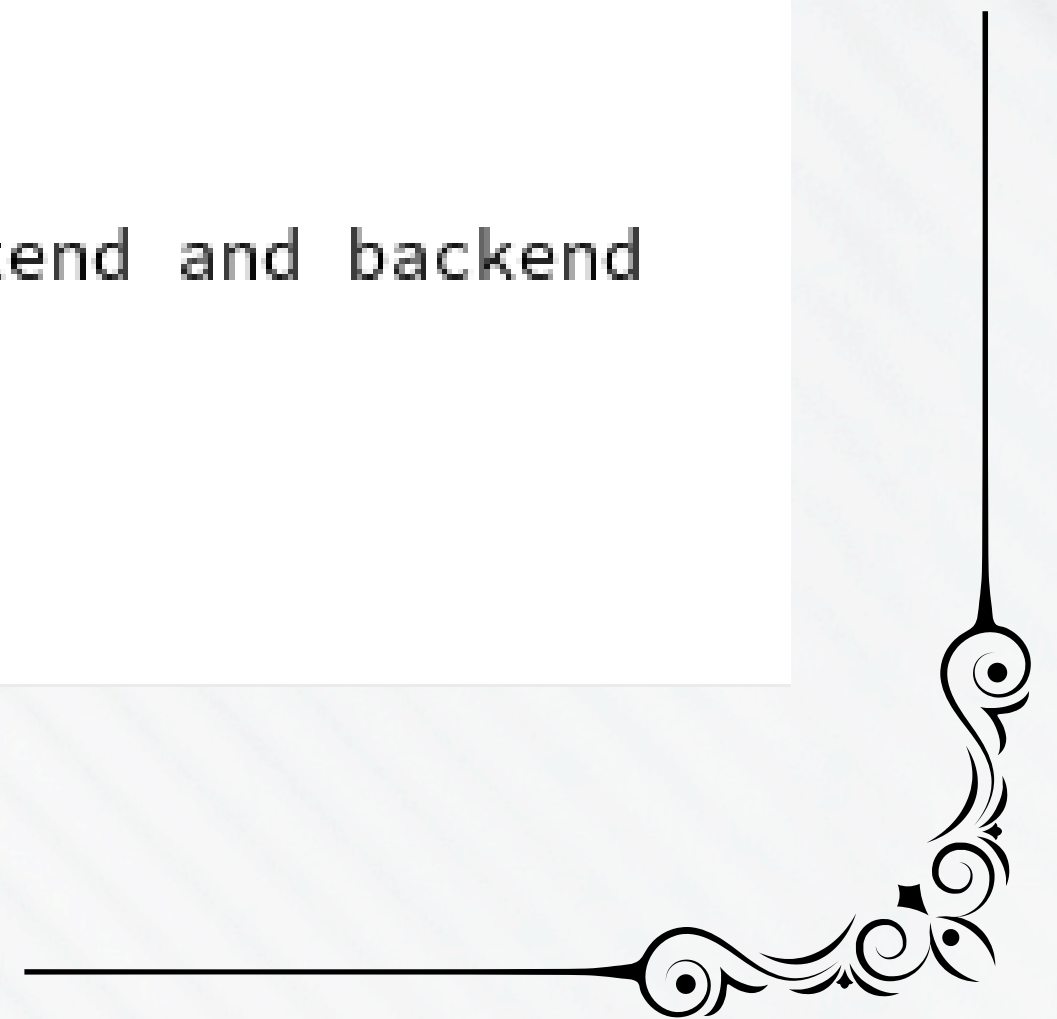
```
1 simple-webapp/
2 |— docker-compose.yaml      ← wires all 5 services together
3 |— frontend/
4 |   |— index.html          ← calls /api/hello and /api/db-check
5 |   |— Dockerfile
6 |— backend/
7 |   |— app.py              ← Flask API + /metrics for Prometheus
8 |   |— requirements.txt
9 |   |— Dockerfile
10 |— proxy/
11 |   |— nginx.conf          ← routes / → frontend, /api → backend
12 |   |— Dockerfile
13 |— monitoring/
14 |   |— prometheus.yml      ← scrapes backend metrics every 5s
15
```

FILE STRUCTURE




```
17
18 frontend-net:  frontend ]
19                                     — proxy (on both)
20 backend-net:  backend ]
21                                     |
22                                     | — can reach both frontend and backend
23                                     |
24                                     | db
25                                     |
26                                     | prometheus
27                                     |
28                                     | grafana
```

NETWORKS STRUCTURE




```
26
27 Browser
28   ↓
29   ▼
30 [ proxy ] :8080 ← ONLY service with a published port – the single gate
31   └─ /      → frontend  (no published port)
32   └─ /api/   → backend   (no published port)
33
34       └─ GET  /api/items → reads from Postgres
35       └─ POST /api/items → writes to Postgres
36
37           ↓
38       [ db ] (no published port)
39       Postgres data → webapp_db_data (named volume)
```

ARCHITECTURE

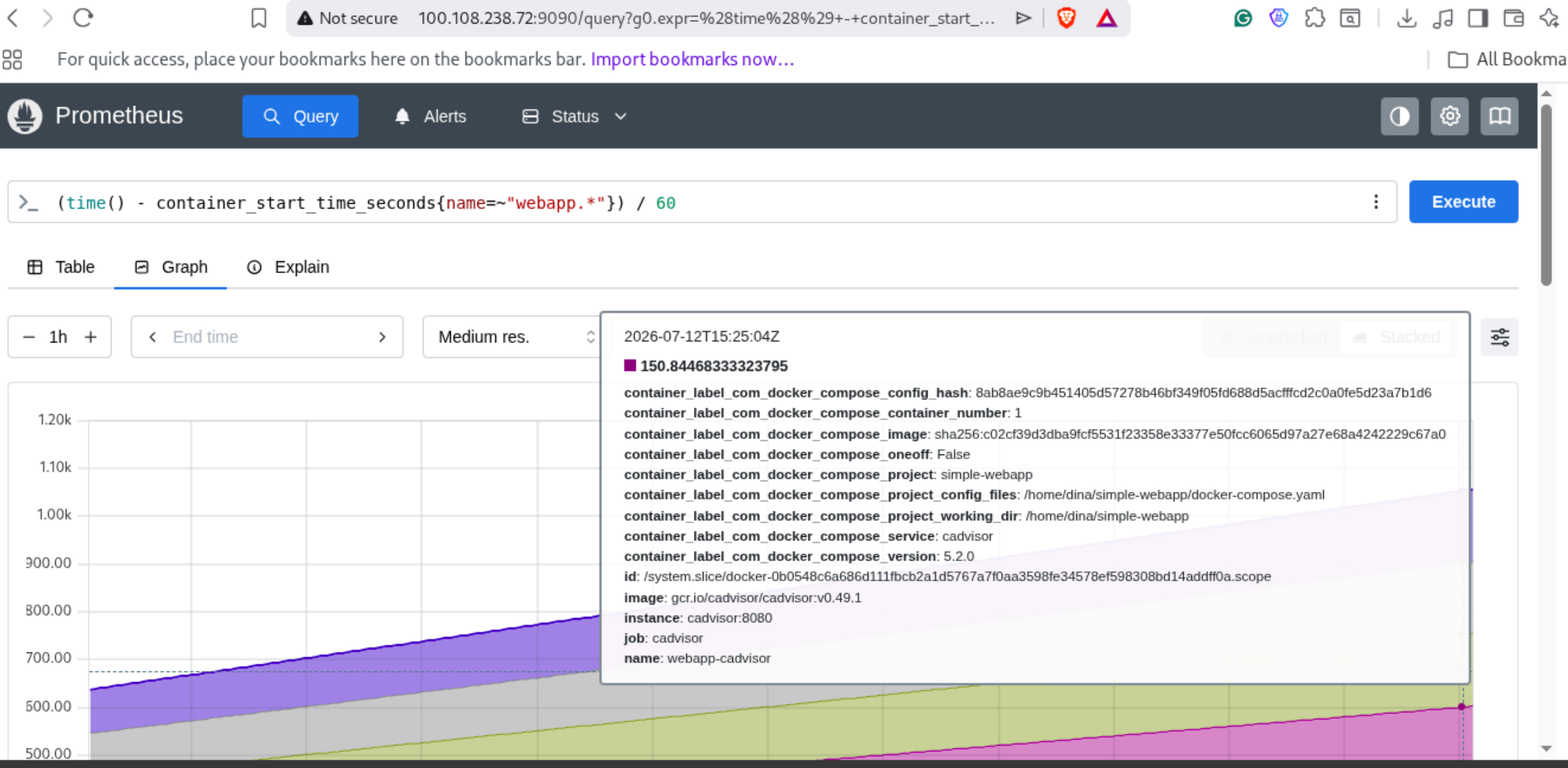
\$ docker ps

IMAGE	COMMAND	CREATED	STATUS	PORTS
grafana/grafana:latest	"/run.sh"	2 hours ago	Up About an hour (healthy)	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
simple-webapp-proxy	"/docker-entrypoint...."	2 hours ago	Up About an hour (unhealthy)	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
prom/prometheus:latest	"/bin/prometheus --c..."	2 hours ago	Up About an hour (healthy)	0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
simple-webapp-backend	"gunicorn --bind 0.0..."	2 hours ago	Up About an hour (healthy)	5000/tcp
gcr.io/cadvisor/cadvisor:v0.49.1	"/usr/bin/cadvisor -..."	2 hours ago	Up About an hour (healthy)	0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp
postgres:16	"docker-entrypoint.s..."	2 hours ago	Up About an hour (healthy)	5432/tcp
simple-webapp-frontend	"/docker-entrypoint...."	2 hours ago	Up About an hour (healthy)	80/tcp

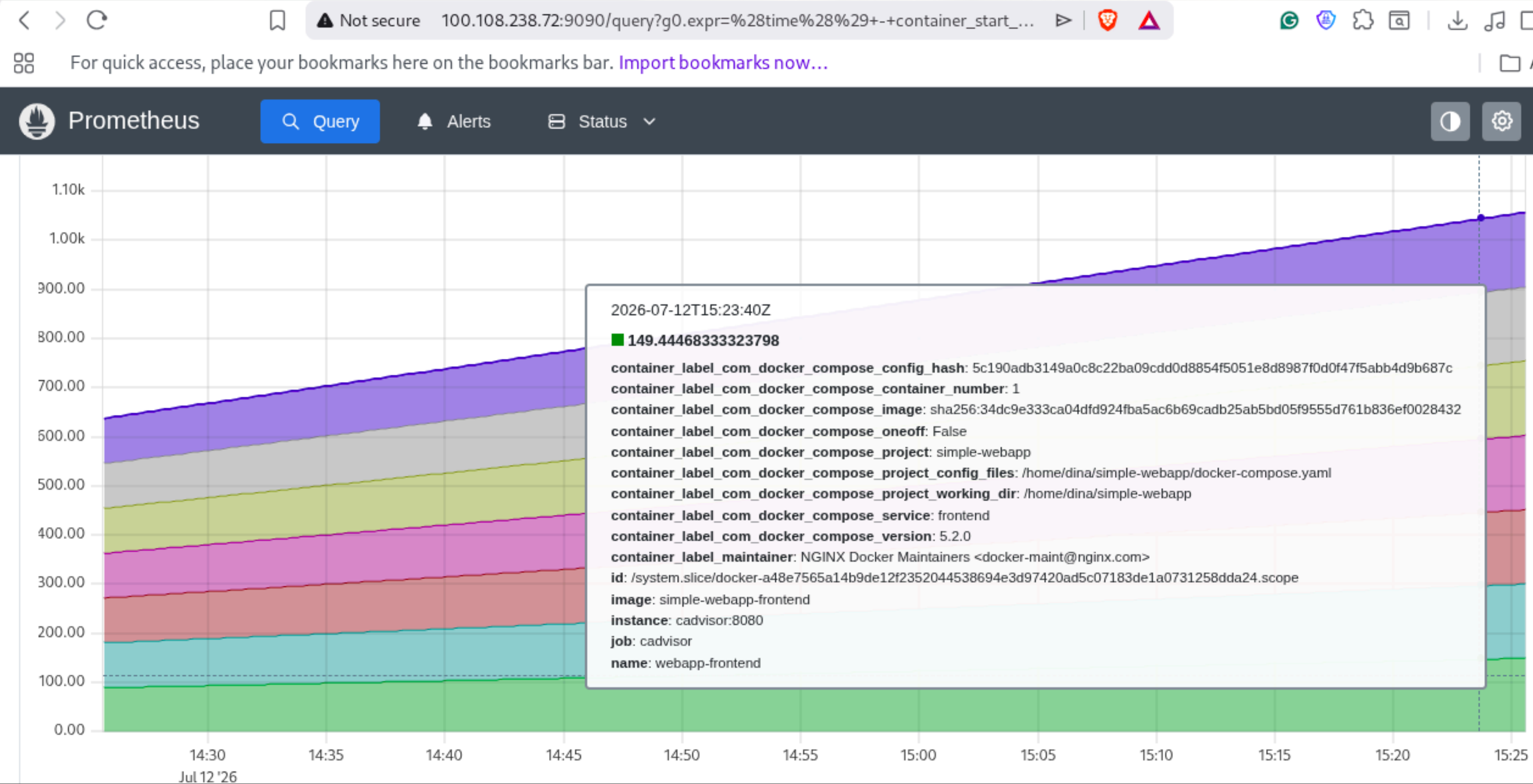
docker compose sample

```
97 # This is the ONLY service with a port published to the host.
98 # All browser traffic must pass through here: / -> frontend, /api -> backend.
99 proxy:
100   build: ./proxy
101   container_name: webapp-proxy
102   restart: unless-stopped
103   deploy:
104     resources:
105       limits:
106         cpus: "0.25"
107         memory: 128M
108   logging:
109     driver: json-file
110     options:
111       max-size: "10m"
112       max-file: "3"
113   healthcheck:
114     test: ["CMD", "wget", "--spider", "-q", "http://localhost:80/"]
115     interval: 10s
116     timeout: 5s
117     retries: 5
118     start_period: 5s
119   ports:
120     - "8080:80"
121   depends_on:
122     frontend:
123       condition: service_healthy
124     backend:
125       condition: service_healthy
126   networks:
127     - frontend-net
128     - backend-net
```


Prometheus



Prometheus



Prometheus

Browser address bar: [100.108.238.72:9090/query?g0.expr=%28time%28%29+-+container_start_...](#)

For quick access, place your bookmarks here on the bookmarks bar. [Import bookmarks now...](#)

Prometheus Query Alerts Status

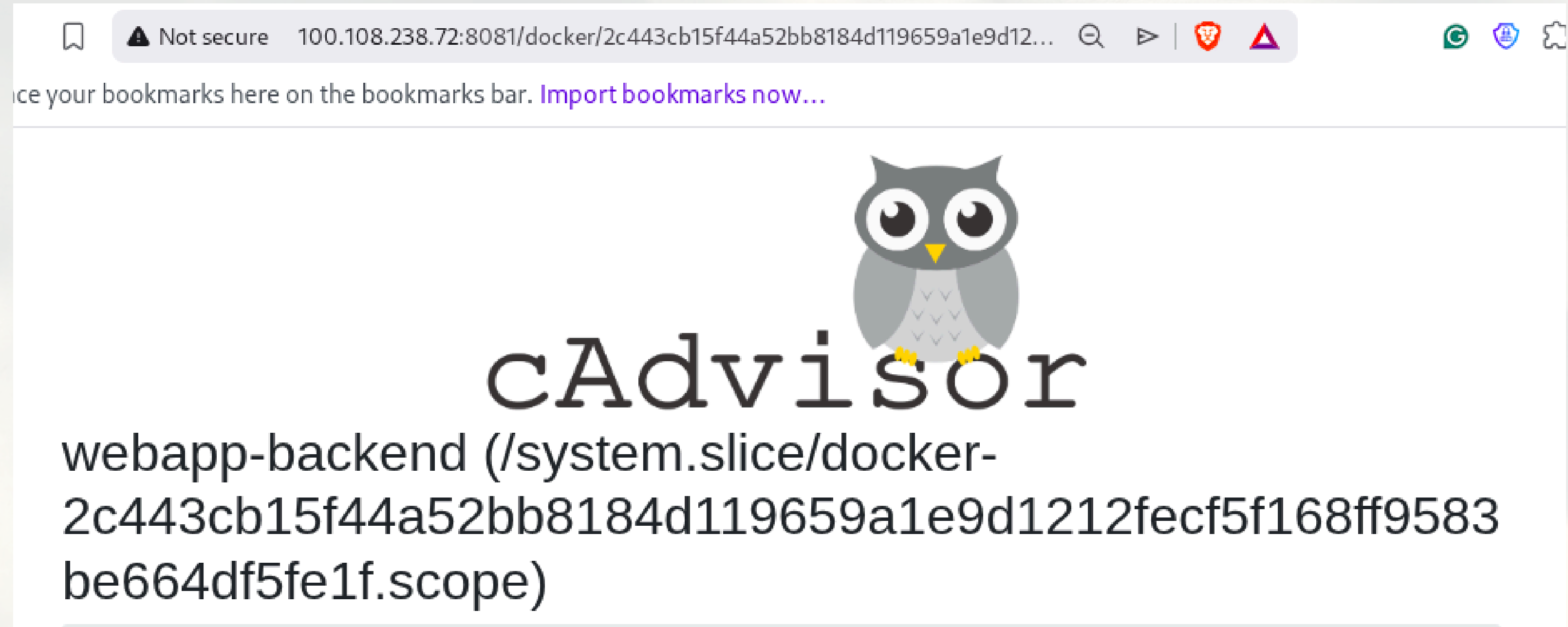
```
3ba2661e2c4dcf86f89d1daf732037753e2332f14421d924394ca1862c240fc4.scope", image="grafana/grafana:latest", instance="cadvisor:8080", job="cadvisor", name="webapp-grafana"}

{container_label_com_docker_compose_config_hash="8ab8ae9c9b451405d57278b46bf349f05fd688d5acffcd2c0a0fe5d23a7b1d6", container_label_com_docker_compose_container_number="1",
container_label_com_docker_compose_image="sha256:c02cf39d3dba9fcf5531f23358e33377e50fcc6065d97a27e68a4242229c67a0", container_label_com_docker_compose_oneoff="False",
container_label_com_docker_compose_project="simple-webapp", container_label_com_docker_compose_project_config_files="/home/dina/simple-webapp/docker-compose.yaml",
container_label_com_docker_compose_project_working_dir="/home/dina/simple-webapp", container_label_com_docker_compose_service="cadvisor", container_label_com_docker_compose_version="5.2.0",
id="/system.slice/docker-0b0548c6a686d111fbc2a1d5767a7f0aa3598fe34578ef598308bd14adff0a.scope", image="gcr.io/cadvisor/cadvisor:v0.49.1", instance="cadvisor:8080", job="cadvisor", name="webapp-cadvisor"}

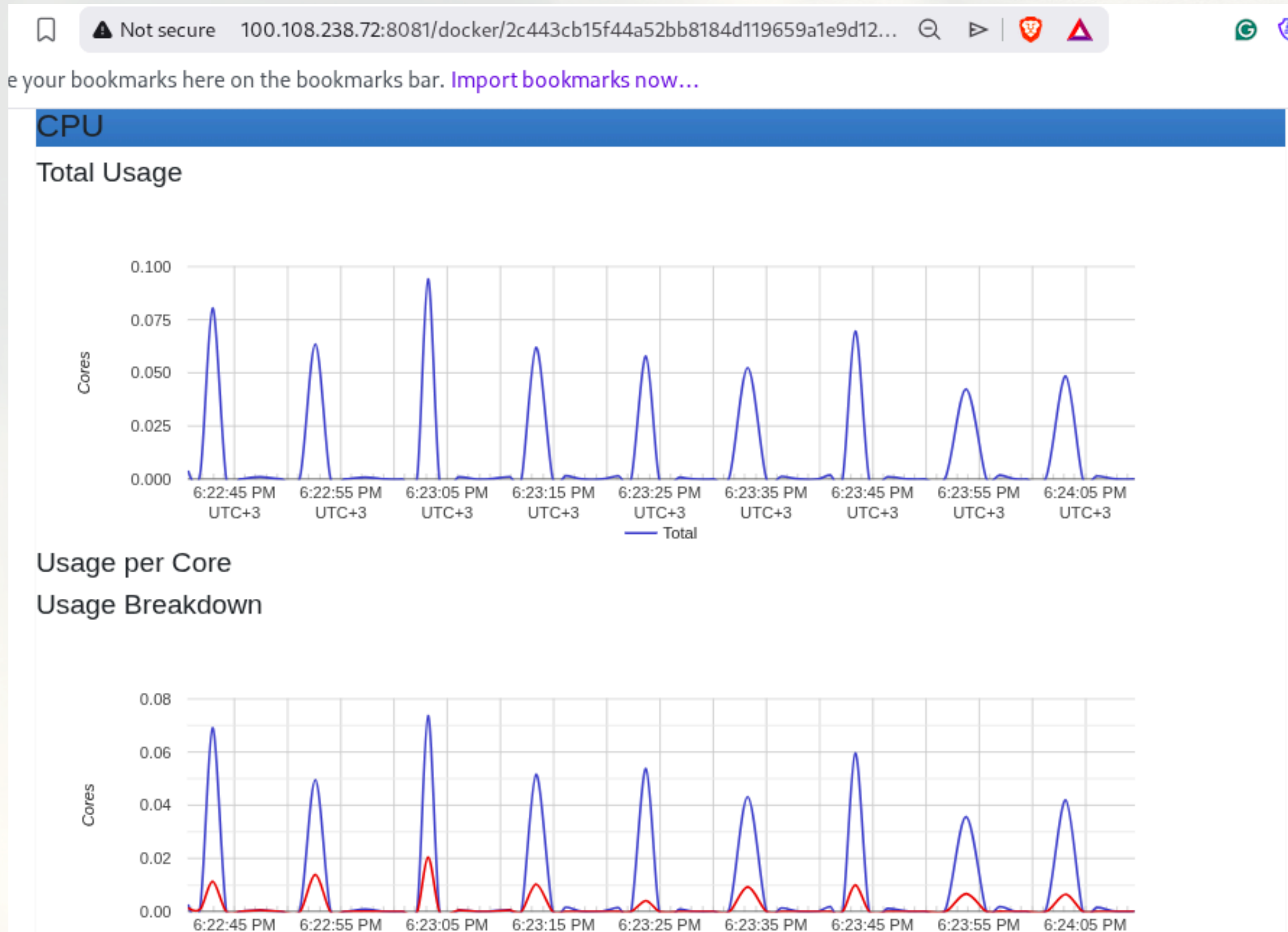
{container_label_com_docker_compose_config_hash="92eb92311e6b983f604ffe318eb6daeee4bd75affe92e86598297107eaccaef29", container_label_com_docker_compose_container_number="1",
container_label_com_docker_compose_image="sha256:44cbf408c5417a24daf1b1ec94625f34251ee99af8c3a51e73900a8769457038", container_label_com_docker_compose_oneoff="False",
container_label_com_docker_compose_project="simple-webapp", container_label_com_docker_compose_project_config_files="/home/dina/simple-webapp/docker-compose.yaml",
container_label_com_docker_compose_project_working_dir="/home/dina/simple-webapp", container_label_com_docker_compose_service="db", container_label_com_docker_compose_version="5.2.0",
id="/system.slice/docker-150f089337737fa1c67dd67b73cc549a75169fc1f604e2bb70ea3e74d90ea6f7.scope", image="postgres:16", instance="cadvisor:8080", job="cadvisor", name="webapp-db"}

{container_label_com_docker_compose_config_hash="c7f4b1bada845f988de2399936056bea37a11303b0eafe5ed4379e747c6eb29f", container_label_com_docker_compose_container_number="1",
container_label_com_docker_compose_depends_on="backend:service_healthy:false", container_label_com_docker_compose_image="sha256:9c3c26cff425955a92cc513ac4ad058923ba7659496cad9f7e2f871a01dfdb1f",
container_label_com_docker_compose_oneoff="False", container_label_com_docker_compose_project="simple-webapp", container_label_com_docker_compose_project_config_files="/home/dina/simple-webapp/docker-compose.yaml", container_label_com_docker_compose_project_working_dir="/home/dina/simple-webapp", container_label_com_docker_compose_service="prometheus",
container_label_com_docker_compose_version="5.2.0", container_label_io_prometheus_image_variant="busybox", container_label_maintainer="The Prometheus Authors <prometheus-developers@googlegroups.com>",
container_label_org_opencontainers_image_authors="The Prometheus Authors", container_label_org_opencontainers_image_description="The Prometheus monitoring system and time series database",
container_label_org_opencontainers_image_documentation="https://prometheus.io/docs", container_label_org_opencontainers_image_licenses="Apache License 2.0",
container_label_org_opencontainers_image_source="https://github.com/prometheus/prometheus", container_label_org_opencontainers_image_title="Prometheus",
container_label_org_opencontainers_image_url="https://github.com/prometheus/prometheus", container_label_org_opencontainers_image_vendor="Prometheus", id="/system.slice/docker-f0f5510cc2aadab035387fed94cc0e4123da2d9aeb401e480c3eaa110f8a347e.scope", image="prom/prometheus:latest", instance="cadvisor:8080", job="cadvisor", name="webapp-prometheus"}
```

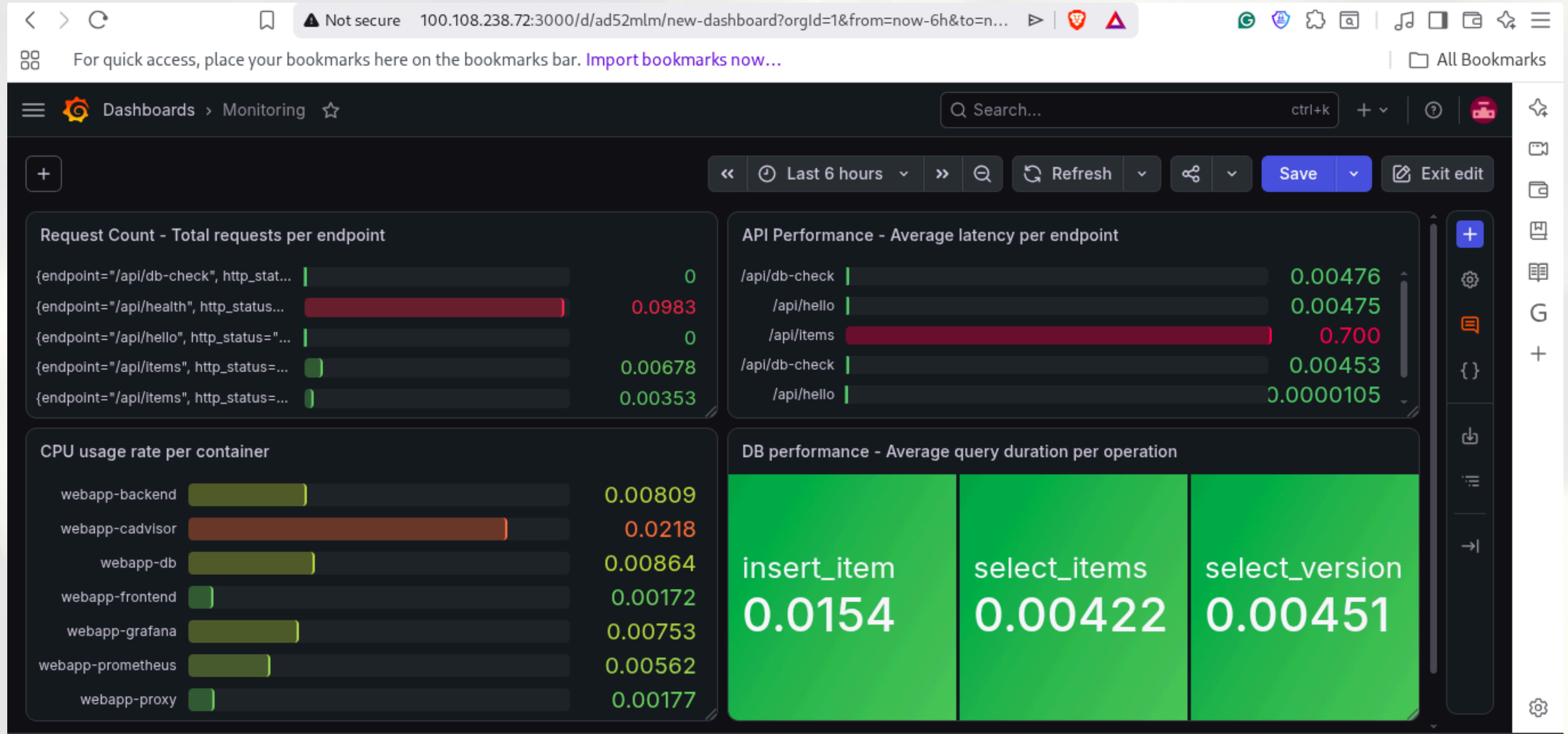

cAdvisor



cAdvisor



Grafana



THANK YOU SO MUCH

 [Github](#)